



SRI KRISHNA INSTITUTE OF TECHNOLOGY

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#57, Chimney Hills, Hesaraghatta Main Road, Chikkabanavara Post, Bengaluru- 560090

Department of Electronics & Communications Engg/VLSI.

Academic Year	:	2025-26	Course Name	:	Fundamentals of Operating Systems
Semester	:	6	Course Code	:	BCS654B
Scheme	:	2022	L: T: P: C	:	3:0:0:3
Total Contact hours	:	40	CIE Marks	:	50
Course Plan Author	:	Dr. Sandeep V. M.	SEE Marks	:	50
Date	:	21 January 2026	Total Marks	:	100

Course Prerequisites:

- Digital Electronics
- Microprocessors/Microcontrollers

Learning Objectives:

- To demonstrate the need and different types of OS
- To discuss suitable techniques for management of different resources
- To analyze different memory, storage, and file system strategies

Course Outcomes:

CO	At the end of the course, student should be able to . . .	Blooms' Level
CO1	: Explain the fundamental concepts of operating systems, including OS objectives, structures, system calls, and services.	L2
CO2	: Analyze process management concepts such as process states, CPU scheduling, inter-process communication, and synchronization mechanisms to handle concurrency issues.	L3
CO3	: Apply memory management techniques and file system concepts, including allocation strategies and directory structures, to solve practical computing problems.	L3

Blooms' Taxonomy:

L1	L2	L3	L4	L5	L6
Remembering	Understanding	Applying	Analyzing	Evaluating	Creating



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Program Outcomes:

PO1	:	Engineering knowledge	PO7	:	Environment and sustainability
PO2	:	Problem analysis	PO8	:	Ethics
PO3	:	Design/development of solutions	PO9	:	Individual and team work
PO4	:	Conduct investigations of complex problems	PO10	:	Communication
PO5	:	Modern tool usage	PO11	:	Project management and finance
PO6	:	The engineer and society	PO12	:	Life-long learning

Program Specific Outcomes:

PSO1:	Graduates will possess advanced knowledge and practical skills in VLSI design including the use of state-of-the-art tools and techniques to develop, analyze and optimize analog, digital and mixed mode systems for real-world applications.
PSO2:	Graduates will demonstrate the ability to conduct interdisciplinary research in VLSI technologies contributing to innovative solutions that address global challenges while advancing sustainable and inclusive technology.
PSO3:	Graduates will be equipped with the entrepreneurial mindset and leadership qualities necessary to create innovative solutions in VLSI design, promoting new ventures, generating employment and taking on roles that foster industry collaboration and societal impact.

CO-PO-PSO Mapping:

CO	Program Outcomes														
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO12	PSO 1	PSO 2	PSO 3
CO1	3	1	-	-	-	-	-	-	-	-	-	1	1	-	-
CO2	2	3	1	-	-	-	-	-	-	-	-	2	2	1	-
CO3	2	2	3	-	-	-	-	-	-	-	-	2	2	2	1

Course Content (Syllabus)

Module 1	CH
Introduction: What operating systems do; Computer System organization; Computer System Organization, Computer System architecture; Operating System operations; Resource Management Operating System Structures: Operating System Services, User and Operating System interface; System calls, Application Program Interface, Types of system calls; Textbook 1: Chapter 1: 1.1, 1.2, 1.3,1.4, 1.5 Chapter 2: 2.1, 2.2 (2.2.1, 2.2.2), 2.3 (2.3.2, 2.3.3)	08



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RBT: L1, L2	
Module 2	
Process Management: Process concept; Process scheduling; Operations on processes; Interprocess Communication Multi-threaded Programming: Overview; Multithreading models, Thread Libraries Textbook 1: Chapter 3: 3.1-3.4, Chapter 4: 4.1, 4.3 5, 4.4	08
RBT: L3	
Module 3	
CPU Scheduling: Basic Concepts, Scheduling criteria, Scheduling algorithms, Thread Scheduling, Process Synchronization: Synchronization: The critical section problem; Peterson's solution; Semaphores; Classical problems of synchronization; Textbook 1: Chapter 5: 5.1, 5.2, 5.3.1, 5.3.2, 5.3.3, 5.3.4, 5.4 Chapter 6: 6.1, 6.2, 6.3, 6.6	08
RBT: L3	
Module 4	
Deadlocks: System model; Deadlock characterization; Methods for handling deadlocks; Deadlock prevention; Deadlock avoidance; Deadlock detection and recovery from deadlock. Memory Management: Background; Contiguous memory allocation; Paging; Structure of page table Textbook 1: Chapter 8: 8.1-8.8 Textbook 1: Chapter 9: 9.1-9.4 (9.4.1, 9.4.2)	08
RBT: L1, L2	
Module 5	
Virtual Memory Management: Background; Demand paging; Copy-on-write; Page replacement; File System Interface: File concept; Access methods; Directory Structure, Protection, File System Implementation: File System Structure, File System Operations, File System Internals: File Systems, File System Mounting; Partition and Mounting, File sharing; Textbook 1: Chapter 10: 10.1-10.3, 10.4 (10.4.1, 10.4.2, 10.4.4.) Chapter 13: 13.1, 13.2, 13.3 (13.3.1, 13.3.2, 13.3.3), 13.4 (13.4.1, 13.4.2) Chapter 15: 15.1-15.4	08
RBT: L1, L2	



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Schedule of Instruction:

Class No	Topic	Date	RB T	C O	Mod e
Module-1					
Introduction, Operating System Structures					
1.	Introduction: What operating systems do;		L1	CO 1	L
2.	Computer System organization; Computer System Organization, Computer System architecture;		L1	CO 1	L
3.	Operating System operations;		L2	CO 1	L
4.	Resource Management		L2	CO 1	L
5.	Operating System Structures: Operating System Services,		L2	CO 1	L
6.	User and Operating System interface		L2	CO 1	L
7.	System calls, Types of system calls;		L2	CO 1	L
8.	Application Program Interface,		L2	CO 1	L
9.	Class Test				
Module-2					
Process Management, Multi-threaded Programming					
10.	Process Management: Process concept;		L2	CO 1	L
11.	Process scheduling;		L3	CO 2	L
12.	Process scheduling;		L3	CO 2	L
13.	Operations on processes;		L2	CO 2	L
14.	Interprocess Communication		L2	CO 2	L
15.	Multi-threaded Programming: Overview		L2	CO 2	L
16.	; Multithreading models,		L2	CO 2	L



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Class No	Topic	Date	RB T	C O	Mode
17.	Thread Libraries		L2	CO 2	L
18.	Class Test				
Module-3					
CPU Scheduling, Process Synchronization					
19.	CPU Scheduling: Basic Concepts,		L2	CO 1	L
20.	Scheduling criteria,		L2	CO 2	L
21.	, Scheduling algorithms		L3	CO 2	L
22.	Thread Scheduling,		L2	CO 2	L
23.	Process Synchronization: Synchronization:		L3	CO 2	L
24.	The critical section problem; Peterson's solution;		L3	CO 2	L
25.	Semaphores;		L3	CO 2	L
26.	Classical problems of synchronization;		L3	CO 2	L
27.	Class Test				
Module-4					
Deadlocks, Memory Management					
28.	Deadlocks: System model;		L3	CO 2	L
29.	Deadlock characterization; Methods for handling deadlocks;		L3	CO 2	L
30.	Deadlock prevention;		L3	CO 2	L
31.	Deadlock avoidance;		L3	CO 2	L
32.	Deadlock detection and recovery from deadlock.		L3	CO 2	L
33.	Memory Management: Background;		L3	CO 3	L



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Class No	Topic	Date	RB T	C O	Mod e
34.	Contiguous memory allocation;		L3	CO 3	L
35.	Paging; Structure of page table		L3	CO 3	L
36.	Class Test				
Module-5					
Virtual Memory Management, File System Interface, File System Internals					
37.	Virtual Memory Management: Background; Demand paging;		L2	CO 3	L
38.	Copy-on-write;		L2	CO 3	L
39.	Page replacement;		L3	CO 3	L
40.	Page replacement		L3	CO 3	L
41.	File System Interface: File concept; Access methods;		L2	CO 1	L
42.	Directory Structure, Protection, File System Implementation: File System Structure, File System Operations,		L2	CO 3	L
43.	File System Internals: File Systems, File System Mounting;		L2	CO 3	L
44.	Partition and Mounting, File sharing;		L2	CO 3	L
45.	Class Test				

Textbooks:

T1	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne, Operating System Principles 10th edition, Wiley-India, 2015
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Reference books:

R1	Ann McHoes Ida M Fylnn, Understanding Operating System, Cengage Learning, 6th Edition, 2010
R2	D.M Dhamdhere, Operating Systems: A Concept Based Approach 3rd Ed, McGraw-Hill, 2013, P.C.P. Bhatt, An Introduction to Operating Systems: Concepts and Practice



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	4th Edition, PHI(EEE), 2014.
R3	William Stallings Operating Systems: Internals and Design Principles, 6th Edition, Pearson, 2008

Web links and Video Lectures (e-Resources):

1.	https://archive.nptel.ac.in/courses/106/105/106105214/
2.	https://archive.nptel.ac.in/courses/106/102/106102132
3.	https://www.youtube.com/playlist?list=PLCHuhFB1N-Ckl_QECkatV212kKUxTUdb3



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Assessment Schedule:

S.N	Assessment Type	Content	CO	Duration	Marks	Date
1.	CIE Test 1	M1,M2,M3	1,2	90 min	25	
2.	CIE Test 2	M3,M4,M5	2,3	90 min	25	
3.	Module Tests	M1,M2,M3,M4,M5	1,2,3		25	

RB – Text Book/Reference Book, ***L** – Lecture, **V**- Videos or any other mode, ***RBT** – Revised Blooms' Taxonomy, **L: T:**
P: C – Theory/Lecture: Tutorial: Practical/Drawing: Credits, **SEE**: Semester End Examination, **CIE**: Continuous Internal Evaluation, **Seminar**: Group of 6-8 students, **Module** 1,2,3,4 & 5,

****The sum of total marks of three tests, two assignments, and seminar will be out of 100 marks and will be scaled down to 50 marks. (As per the scheme), CIE + SEE = 50 + 50 = 100 marks**

Faculty In charge

Course Coordinator

HoD